Robust Equalization for Single-Carrier Underwater Acoustic Communications Based on Parameterized Interference Model

Wei Ge, Zhaohui Wang, *Member, IEEE*, Jingwei Yin, *Member, IEEE*, and Xiao Han, *Member, IEEE*

***Abstract*—There are several sources of artificial and naturally generated impulsive interference in marine environment which could severely degrade the acoustic communication performances. This letter tackles the equalization for single-carrier modulated systems with impulsive interference based on parameterized model, and a two-step equalization algorithm is proposed. In the first step, an iterative interference impulse estimation/cancellation can be performed and initial symbol estimation is obtained by conventional decision feedback equalization. In the second step, a joint channel, impulse, and symbol estimation for single-carrier modulation is applied to enhance the performance. The proposed algorithm is evaluated using partial-band partial-block-duration interference. Simulation results show that the proposed joint estimation algorithm has significant performance advantages over conventional algorithm in various signal-to-interference ratio levels.**

***Index Terms*—Impulsive interference, single carrier, underwater acoustics, joint estimation.**

I. Introduction

UNDERWATER environments are rich in natural and artificial sound sources which can create impulsive interference in underwater acoustic (UWA) communications. Such interference from malicious jamming, sonar operations, natural environments [1], [2] etc. often occurs during the UWA communications, which leads to significant degradation of data transmission [3], [4], [5].

Various interference suppression methods have been investigated for UWA communications. One commonly method is to first detect the samples that are possibly contaminated by interference in the received signal through a threshold testing. And use clipping or nonlinear blanking to mitigate the effect in the interference-pollution samples. Both [6] and [7] detected the positions of possible impulsive noise by a power threshold. Then, [6] measured the impulse amplitude in null subcarriers of orthogonal frequency division multiplexing (OFDM) using the least square (LS) method. A joint channel estimation and impulsive interference mitigation method based on compressed-sensing (CS) was proposed using pilot subcarriers [7]. Another sparse-based method is sparse Bayesian learning (SBL). Considering the fast time-varying characteristics of UWA channel and impulsive interference, [8] developed an joint estimation algorithm to track the UWA channel and impulsive interference by fusing the SBL method and forwad-backward Kalman filtering.

These algorithms above are mainly proposed for OFDM systems. This letter explores general impulsive interference cancellation for single-carrier systems in the time domain. The general interference is represented as partial-band partial-block-duration interference [9], and some other interference like single frequency pulse can be regarded as a special form of this interference. In our previous work [10], the general interference waveform was parameterized with a finite number of unknowns for single-carrier signals. In this letter, we develop a two-step equalization algorithm based on the parameterized model to mitigate the effect of interference. For the first step, an iterative interference impulse estimation/cancellation will be performed and initial symbol estimation is obtained by conventional decision feedback equalization (DFE). In the second step, a joint channel, impulse, and symbol estimation for single-carrier modulation is applied to enhance the performance. In the SBL framework, channel, parameterized impulsive interference, and symbol can be estimated at the same expectation-maximization (EM) iteration. The processing gain is not only obtained from the sparsity prior of the UWA channel but also from the joint processing, which improves the match between the symbol estimation results and the channel and interference. Simulation results show that the proposed joint estimation algorithm has significant performance advantages over conventional algorithm in various signal-to-interference ratio (SIR) levels.

II. System Model

In single-carrier system, the transmitted frame of baseband signal can be written as

$$s(t) = \sum_{n=0}^{N_s - 1} s[n] g(t - nT_s), \qquad t \in [0, T], \tag{1}$$

where $g(t)$ is the pulse shaping function, $\{s[n]\}$ denotes the symbol sequence of length N_s, T_s is the symbol period, and $T := N_s T_s$ is the signal duration.

Manuscript received 18 October 2022; revised 7 November 2022; accepted 16 November 2022. Date of publication 2 December 2022; date of current version 11 September 2024. This work was supported in part by the National Science Fund for Distinguished Young Scholars under Grant 62125104; in part by the National Key Research and Development Program of China under Grant 2021YFC2801204; in part by the National Natural Science Foundation of China under Grant 62127801; and in part by the Foundation of the Acoustic science and Technology Laboratory. The associate editor coordinating the review of this article and approving it for publication was N. Saeed. *(Corresponding author: Jingwei Yin.)*

Wei Ge and Jingwei Yin are with the Qingdao Innovation and Development Center, Harbin Engineering University, Qingdao 266400, China (e-mail: gewei@hrbeu.edu.cn; yinjingwei@hrbeu.edu.cn).

Zhaohui Wang is with the Department of Electrical and Computer Engineering, Michigan Technological University, Houghton, MI 49931 USA (e-mail: zhaohuiw@mtu.edu).

Xiao Han is with the Acoustic Science and Technology Laboratory and the Key Laboratory of Marine Information Acquisition and Security, Ministry of Industry and Information Technology, Harbin Engineering University, Harbin 150001, China (e-mail: hanxiao1322@hrbeu.edu.cn).

Digital Object Identifier 10.1109/LWC.2022.3223533

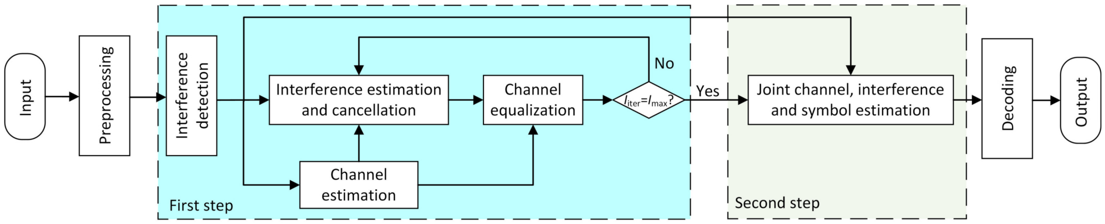

Fig. 1. Illustration of the block diagram for robust equalization. I_{iter} and $I_{\max}$ indicate the index and maximum index of the iteration, respectively.

Considering an approximately time invariant channel model

$$h(\tau) = \sum_{p=1}^{N_{\text{pa}}} A_p \delta(\tau - \tau_p), \tag{2}$$

where N_{pa} is the number of arriving paths, τ_p and A_p are the delay and amplitude of the pth path, respectively. In the environment with impulsive interference, the received baseband signal can be expressed as

$$y(t) = \sum_{n=1}^{N_{\text{s}}-1} \sum_{p=1}^{N_{\text{pa}}} A_p s[n] g(t - \tau_p - nT_{\text{s}}) + I(t) + w(t), \tag{3}$$

where $I(t)$ represents the impulsive interference and $w(t)$ is the baseband noise. Let f_{sB} indicates the baseband sampling rate. The discrete form of Eq. (3) can be represented as

$$y[m] = \sum_{n=1}^{N_{\text{s}}-1} \sum_{p=1}^{N_{\text{pa}}} A_p s[n] g\left(\frac{m}{f_{\text{sB}}} - \tau_p - nT_{\text{s}}\right) + I[m] + w[m], \tag{4}$$

$$= \sum_{l=0}^{L-1} s[m-l] h_l + I[m] + w[m], \tag{5}$$

where $\{h_l\}$ is the discrete channel taps of length L.

In our previous work [10], the interference $I[m]$ can be described as a parameterized model

$$I[m] = \sum_{\ell=1}^{N_{\text{int}}} p_{m,\ell} e^{j2\pi(f_{\text{cI},\ell} - f_{\text{c}})\frac{m}{f_{\text{sB}}}} \sum_{n=0}^{N_{\text{I},\ell}-1} u_{n,\ell} \varrho_{m,n,\ell}, \tag{6}$$

with

$$p_{m,\ell} := p_{T_{\text{I},\ell}}\left(\frac{m}{f_{\text{sB}}} - \tau_{\text{I},\ell}\right), \qquad u_{n,\ell} := I_\ell\left(\frac{n}{f_{\text{sBI},\ell}}\right),$$
$$\varrho_{m,n,\ell} := \text{sinc}\left[\pi f_{\text{sBI},\ell}\left(\frac{m}{f_{\text{sB}}} - \tau_{\text{I},\ell} - \frac{n}{f_{\text{sBI},\ell}}\right)\right] e^{-j2\pi f_{\text{cI},\ell}\tau_{\text{I},\ell}}.$$

where $p_{m,\ell} = 0$ or 1 indicates if the ℓth interference contributes to the mth sample. I_ℓ is the ℓth interfering waveform, and $\tau_{\text{I},\ell}$ is the arrival time. N_{int} denotes the total number of individual interference, N_{I} represents the total number of interference parameters related to the bandwidth B_{I} and duration T_{I}. f_{sBI} and f_{cI} denotes the baseband sampling rate and center frequency of each individual interference. Stack all the $I\{m\}$ into the vector $\mathbf{i}$ of length N_{fra}, we have

$$\mathbf{i} = \mathbf{\Lambda\Gamma u}, \tag{7}$$

where $\mathbf{\Lambda}$ is a $N_{\text{fra}} \times N_{\text{fra}}$ diagonal matrix composed of $\{e^{j2\pi(f_{\text{cI},\ell} - f_{\text{c}})\frac{m}{f_{\text{sB}}}}\}$ and f_{c} is the center frequency of the single-carrier waveform, $\mathbf{u}$ is a vector of length $N_{\mathbf{u}}$ by stacking $\{u_{n,\ell}\}$, and $\mathbf{\Gamma}$ is a $N_{\text{fra}} \times N_{\mathbf{u}}$ block matrix composed of $\varrho_{m,n,\ell}$, where $N_{\mathbf{u}}$ is the total length of all interference.

Stacking $\{y[m]\}$ and $\{w[m]\}$ into vectors of length N_{fra}, $\{s[n]\}$ into vector of length N_s, a concise form of Eq. (7) can be written as

$$\mathbf{y} = \mathbf{Hs} + \mathbf{\Lambda\Gamma u} + \mathbf{w}, \tag{8}$$

where $\mathbf{H}$ is a Toeplitz channel matrix formed by $\{h_l\}$.

III. Proposed Equalization Algorithm

In this section, we adopt a two-step equalization to mitigate the effect of interference. The first step is iterative interference estimation and cancellation, the main purpose of which is to get an initial estimation of symbols. The second step is joint channel, interference and symbol estimation to improve the performance of the first step. The block diagram of the proposed method is shown in Fig. 1. The processing modules are briefly described in the following.

- *Preprocessing:* The received baseband signal is taken as the input, and the preprocessing module estimates and compensates Doppler which is caused by the mobility of platform or channel dynamics.
- *Interference detection:* A Page test detector [10] for interference detection is applied to detect the presence of an interference waveform and obtain the starting/ending times of the interference.
- *Channel estimation:* The UWA channels are estimated by the received training sequence with considering the impulsive interference after interference detection [10].
- *Interference estimation and cancellation:* With the estimation results of Page test, the interference $\mathbf{u}$ is estimated under LS criteria.If interference exist, the contributions of interference will be estimated and subtracted from the received signal.
- *Channel equalization:* A block-based time-domain DFE can be used for channel equalization in the first step.
- *Joint channel, interference and symbol estimation:* After the iterative processing, a joint sparse channel, interference and symbol estimation based on SBL is performed to improve the equalization performance.

A. Iterative Interference Estimation and Cancellation

In this step, the baseband signal $\mathbf{y}$ is divided into several blocks, which is shown in Fig. 2. Block-based interference

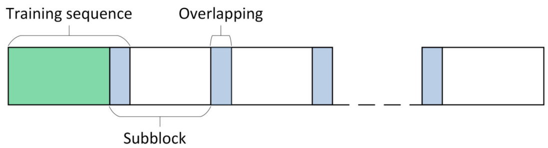

Fig. 2. Illustration of the block-based processing.

cancellation and channel equalization are applied to signal processing. Overlapping will ensure the processing continuity, and the length of the overlapping should be larger than the channel length. We denote $\mathbf{y}_b := (y[1], y[2], \ldots, y[N_b])^{\mathrm{T}}$ as an inter-block-interference(IBI)-free block and $\mathbf{s}_b$ with the corresponding desired symbols. The receiver performs iterative interference estimation and cancellation after detecting the interference waveform. We Denote ℓ as the individual interference index of each block, and $\tau_{\mathrm{I},\ell}$ and $T_{\mathrm{I},\ell}$ as arrival time and duration of the ℓth interference waveform, respectively.

In the ith iteration to process $\mathbf{y}_b$. The desired signal is reconstructed as $\hat{\mathbf{H}}\hat{\mathbf{s}}_b^{(i-1)}$, where $\mathbf{H}$ is estimated by the training sequence and estimated symbols are from the $(i-1)$th iteration. We define the following based on Eq. (8),

$$\mathbf{i}_b := \mathbf{y}_b - \hat{\mathbf{H}}\hat{\mathbf{s}}_b^{(i-1)} = \hat{\mathbf{B}}_b\mathbf{u}_b + \bar{\mathbf{w}}_b, \tag{9}$$

where $\hat{\mathbf{B}}_b := \hat{\boldsymbol{\Lambda}}_b\hat{\boldsymbol{\Gamma}}_b$, $\hat{\boldsymbol{\Lambda}}_b$ and $\hat{\boldsymbol{\Gamma}}_b$ are formed based on the estimations of the parameters $(f_{\mathrm{cI},\ell}, B_{\mathrm{I},\ell}, \tau_{\mathrm{I},\ell})$ (c.f. Eq. (7)). $\bar{\mathbf{w}}_b := (\mathbf{H}\mathbf{s}_b - \hat{\mathbf{H}}\hat{\mathbf{s}}_b^{(i-1)}) + \mathbf{w}_b$ is an equivalent noise composed of the Gaussian noise and estimation error.

Vector $\mathbf{u}_b$ can be estimated by LS criterion,

$$\hat{\mathbf{u}}_b^{(i)} = [\hat{\mathbf{B}}_b^{\mathrm{H}}\hat{\mathbf{B}}_b]^{-1}\hat{\mathbf{B}}_b\mathbf{i}_b. \tag{10}$$

Then, the interference can be reconstructed and subtracted from $\mathbf{y}_b$. We define

$$\mathbf{z}_b^{(i)} := \mathbf{y}_b - \hat{\mathbf{B}}_b\hat{\mathbf{u}}_b^{(i)} = \mathbf{H}\mathbf{s}_b + \check{\mathbf{w}}_b, \tag{11}$$

where $\check{\mathbf{w}}_b := (\mathbf{B}_b\mathbf{u}_b - \hat{\mathbf{B}}_b\hat{\mathbf{u}}_b^{(i)}) + \mathbf{w}_b$ is an equivalent noise term composed of the Gaussian noise and residual interference. The channel equalization (e.g., DFE with phase-locked loop [10]) is then be performed based on the $\mathbf{z}_b$ to get the estimation of $\mathbf{s}_b$.

B. Joint Channel, Interference and Symbol Estimation

We rewrite Eq. (8) as

$$\mathbf{y}_b = \bar{\mathbf{s}}_b\mathbf{h} + \boldsymbol{\Lambda}_b\boldsymbol{\Gamma}_b\mathbf{u}_b + \bar{\mathbf{w}}_b, \tag{12}$$

where $\bar{\mathbf{s}}$ is a matrix formed by desired symbols

$$\bar{\mathbf{s}}_b := \begin{bmatrix} s_b[0] & \cdots & s_b[-L+2] & s_b[-L+1] \\ s_b[1] & \cdots & s_b[-L+3] & s_b[-L+2] \\ \vdots & \ddots & \ddots & \vdots \\ s_b[N_b-1] & \cdots & s_b[N_{\mathrm{b}}-L+1] & s_b[N_b-L] \end{bmatrix}. \tag{13}$$

Here, L is the channel length. $\mathbf{h} := [h_0, h_1, \ldots, h_{L-1}]^{\mathrm{T}}$ is the channel vector.

$\bar{\mathbf{w}}_b$ is assumed as complex Gaussian distribution with variance $1/\sigma^2$ and zero mean. Therefore, the likelihood of Eq. (12) can be expressed as

$$\begin{aligned} &\mathrm{p}(\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b|\mathbf{h}, \sigma^2) \\ &\quad = (\sigma^2/\pi)^{N_b}\exp\{-\sigma^2\|\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b - \bar{\mathbf{s}}_b\mathbf{h}\|^2\}, \end{aligned} \tag{14}$$

where $\|\cdot\|$ indicates the Euclidean norm. $\mathbf{h}$ can be assumed as a complex Gaussian distribution with variance $1/\alpha[j]$ and zero mean for its jth element as

$$\mathrm{p}(\mathbf{h}|\boldsymbol{\alpha}) = \pi^{-L}\prod_{j=1}^{L}\alpha_j\exp\Big(-\mathbf{h}^{\mathrm{H}}\mathrm{diag}(\boldsymbol{\alpha})\mathbf{h}\Big), \tag{15}$$

where $\boldsymbol{\alpha} = (\alpha[1], \alpha[2], \ldots, \alpha[L])^{\mathrm{T}}$. Parameters $\boldsymbol{\alpha}$ and σ^2 are assumed as Gamma distributions,

$$\mathrm{p}(\boldsymbol{\alpha}) = \prod_{i=1}^{L}\gamma(\alpha[i]|a, b), \tag{16}$$

$$\mathrm{p}(\sigma^2) = \gamma(\sigma^2|c, d), \tag{17}$$

where $\gamma(\cdot)$ represents the Gamma distribution with parameters a, b, c, and d.

From Eq. (14) and Eq. (15), it is easy to check that $\mathrm{p}(\mathbf{h}|\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b, \boldsymbol{\alpha}, \sigma^2)$ is complex Gaussian:

$$\begin{aligned} &\mathrm{p}(\mathbf{h}|\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b, \boldsymbol{\alpha}, \sigma^2) \\ &\quad = (\pi)^{-L}|\boldsymbol{\Sigma}|^{-1}\exp\{-(\mathbf{h} - \boldsymbol{\mu})^{\mathrm{H}}\boldsymbol{\Sigma}^{-1}(\mathbf{h} - \boldsymbol{\mu})\}, \end{aligned} \tag{18}$$

where $|\cdot|$ denotes the matrix determinant, and

$$\boldsymbol{\Sigma} = (\sigma^2\bar{\mathbf{s}}^{\mathrm{H}}\bar{\mathbf{s}} + \mathrm{diag}(\boldsymbol{\alpha}))^{-1}, \tag{19}$$

$$\boldsymbol{\mu} = \sigma^2\boldsymbol{\Sigma}\bar{\mathbf{s}}^{\mathrm{H}}(\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b), \tag{20}$$

Then following the LS criterion as Eq. (10), we get the estimation of $\mathbf{u}_b$ as

$$\hat{\mathbf{u}}_b = [\hat{\mathbf{B}}_b^{\mathrm{H}}\hat{\mathbf{B}}_b]^{-1}\hat{\mathbf{B}}_b(\mathbf{y}_b - \bar{\mathbf{s}}_b\boldsymbol{\mu}). \tag{21}$$

Based on Eq. (20), the optimal $\mathbf{h}$ is given by

$$\hat{\mathbf{h}} = \boldsymbol{\mu}, \tag{22}$$

which depends on $\boldsymbol{\alpha}$ and σ^2.

The estimation of $\boldsymbol{\alpha}$ and σ^2 can be obtained by maximizing $\mathrm{p}(\mathbf{h}, \boldsymbol{\alpha}, \sigma^2|\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b)$. Here, we apply the EM algorithm in the framework of SBL. We denote $\mathbf{D} = (\mathbf{s}_b[0], s_b[1], \ldots, s_b[N_b-1])^{\mathrm{T}}$. From Eqs. (14)-(17), the log-likelihood function can be obtained as

$$\mathcal{L} = \ln(\mathrm{p}(\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b|\mathbf{h}, \sigma^2; \mathbf{D})\mathrm{p}(\mathbf{h}|\boldsymbol{\alpha})\mathrm{p}(\boldsymbol{\alpha})\mathrm{p}). \tag{23}$$

The optimal $\mathbf{D}$ can be obtained by maximizing the following equation:

$$E_{\mathrm{p}(\mathbf{h}|\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b, \boldsymbol{\alpha}, \sigma^2)}[\ln(\mathrm{p}(\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b|\mathbf{h}, \sigma^2; \mathbf{D}))], \tag{24}$$

where $E_{\mathrm{p}(\mathbf{h}|\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b, \boldsymbol{\alpha}, \sigma^2)}$ represents the expected value with respect to the distribution in Eq. (18). Eq. (24) is equivalent to solving the problem of

$$\min_{\mathbf{D}}\|\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b - \bar{\mathbf{s}}_b\boldsymbol{\mu}\|^2 + \mathrm{tr}\Big(\bar{\mathbf{s}}_b\boldsymbol{\Sigma}\bar{\mathbf{s}}_b^{\mathrm{H}}\Big) \tag{25}$$

We denote $\mathbf{I}_b := \mathbf{B}_b\mathbf{u}_b$ and $I_b[i]$ as the ith element of $\mathbf{I}_b$. From Eq. (25), the symbols in $\bar{\mathbf{s}}_b$ can be calculated by solving

$$\min_{\bar{\mathbf{s}}_b[i]} |y_b[i] - I_b[i] - \bar{\mathbf{s}}_b[i]\boldsymbol{\mu}|^2 + \mathrm{tr}\left(\bar{\mathbf{s}}_b[i]\boldsymbol{\Sigma}\bar{\mathbf{s}}_b^{\mathrm{H}}[i]\right) \quad (26)$$

where $\bar{\mathbf{s}}_b[i]$ is the ith row of $\bar{\mathbf{s}}_b$. After updating all the elements of $\bar{\mathbf{s}}_b$, we take out the first column as the estimation of $\mathbf{D}$.

By ignoring the independent terms of Eq. (23) relative to $\boldsymbol{\alpha}$, the optimal $\boldsymbol{\alpha}$ can be obtained by maximizing

$$E_{\mathrm{p}(\mathbf{h}|\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b, \boldsymbol{\alpha}, \sigma^2)}[\ln(\mathrm{p}(\mathbf{h}|\boldsymbol{\alpha})\mathrm{p}(\boldsymbol{\alpha}))] \quad (27)$$

Let the derivative of Eq. (27) with respect to $\alpha[j]$ be zero, we get

$$\alpha[j] = \frac{a}{b + \Sigma[j,j] + |\mu[j]|^2}, \ j = 1, \ldots, L \quad (28)$$

where $\mu[j]$ and $\Sigma[j,j]$ denote the jth entry of $\boldsymbol{\mu}$ in Eq. (20) and the (j,j)th entry of $\boldsymbol{\Sigma}$ in Eq. (19), respectively. Also, σ^2 can be optimized by maximizing

$$E_{\mathrm{p}(\mathbf{h}|\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b, \boldsymbol{\alpha}, \sigma^2)}\left[\ln(\mathrm{p}(\mathbf{y}_b - \mathbf{B}_b\mathbf{u}_b|\mathbf{h}, \sigma^2; \mathbf{D})\mathrm{p}(\sigma^2))\right], \quad (29)$$

which leads to

$$\sigma^2 = \frac{c + N_b - 1}{\tilde{d}} \quad (30)$$

where

$$\tilde{d} = d + (\mathbf{y}_b - \mathbf{I}_b)^{\mathrm{H}}(\mathbf{y}_b - \mathbf{I}_b) - (\mathbf{y}_b - \mathbf{I}_b)^{\mathrm{H}}\bar{\mathbf{s}}_b\boldsymbol{\mu} \quad (31)$$
$$- \boldsymbol{\mu}^{\mathrm{H}}\bar{\mathbf{s}}_b^{\mathrm{H}}(\mathbf{y}_b - \mathbf{I}_b) + \mathrm{tr}(\bar{\mathbf{s}}_b\boldsymbol{\Sigma}\bar{\mathbf{s}}_b^{\mathrm{H}}) + \boldsymbol{\mu}^{\mathrm{H}}\bar{\mathbf{s}}_b^{\mathrm{H}}\bar{\mathbf{s}}_b\boldsymbol{\mu} \quad (32)$$

IV. Computational Complexity Analysis

The computational complexity is mainly determined by the substructures of the proposed receiver. The interference is detected by Page test detector, which requires $\mathcal{O}(N_b)$. An algorithm based on linear minimum mean squared error (LMMSE) [10] can be applied for channel estimation, which needs $\mathcal{O}({N_{\mathrm{p}}}^3)$, where N_{p} is the length of training sequence. The cost of interference estimation and cancellation is $\mathcal{O}({N_b}^3)$. DFE algorithm can be implemented to channel equalization and will consume $\mathcal{O}(N_b(P_1+P_2)^2)$, where P_1 and P_2 are the length of DFE taps related to channel delay. The computational complexity of joint channel, interference and symbol estimation is $\mathcal{O}({N_b}^3)$.

V. Simulation Results

Two kinds of UWA channel are used in the simulation. The first one is a spare UWA channel with three discrete paths, where the delay between two paths follows an exponential distribution with mean of 10 ms. The amplitudes are Rayleigh distributed with an average power that decreases exponentially with the delay. The other one is a real UWA channel with two arrival paths, and the time delay between the paths is about 5 ms. The communications data was collected in an experiment held in Bohai Gulf, China.

The center frequency of the single-carrier signal is 6 kHz and the bandwidth is 2 kHz. There are 1000 symbols in the simulated channels and 7500 symbols in the experiment, including 500 training symbols for each frame.The data symbols are modulated by interleaving and 1/2 rate convolution-encoded quadrature phase-shift keying (QPSK) constellations. The simulated interference is the partial-band partial-block-duration interference in [9], which can be generated by passing white Gaussian noise through a bandpass filter according to the parameters of center frequency $f_{\mathrm{cI}} =$ 6.35 kHz, duration $T_{\mathrm{I}} = 30$ ms, and bandwidth $B_{\mathrm{I}} = 500$ Hz. The block-based processing is applied in this section, and the length of each subblock is 500 symbols.

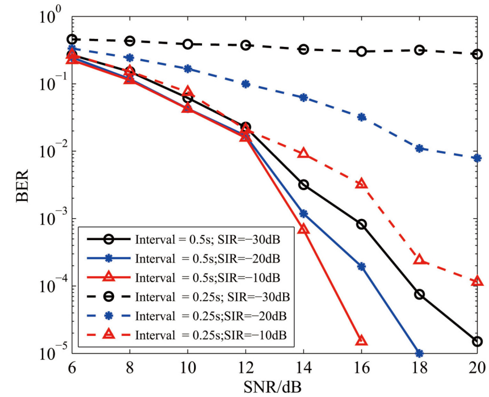

Fig. 3. The BER performance of JCISE algorithm at different SIR levels in the simulated channels.

Through this letter, we define the SIR and signal-to-noise ratio (SNR) as follow:

$$\mathrm{SIR} = \frac{P_{\mathrm{s}}}{P_{\mathrm{I}}}, \mathrm{SNR} = \frac{P_{\mathrm{s}}}{N_0}, \quad (33)$$

where P_{s} is the average power of the single-carrier signal in the communication band, P_{I} is the average interference band, and N_0 is the variance of the ambient noise.

In this section, we compare the performance of three equalization algorithm. The first one is a conventional direct adaptive DFE (DADFE) with recursive least square algorithm. The second one is the iterative interference estimation and cancellation (IIEC) algorithm in Section III-A. The third one is joint channel, interference and symbols estimation (JCISE) algorithm. The interference is detected by a Page test detector in [10].

Fig. 3 shows the bit error rate (BER) performance of the JCISE algorithm at different SIR levels in the simulated channels. The interference interval indicates the intensity of the individual partial-band partial-block-duration interference. That is, the interval of two interference impulses is 0.5 or 0.25 s. The BER performance is affected by the intensity and a smaller density leads to a better performance. In general, a lower SIR gives better performance.

Figs. 4 and 5 show the BER performance for the three algorithms at different SIR levels in the simulated channels. For interference-free scenarios, although no interference can be detected by the Page test, the performance of JCISE algorithm is better than the conventional DADFE algorithm, where the processing gain is from the joint estimation. In the interference-present scenario, the performance of IIEC

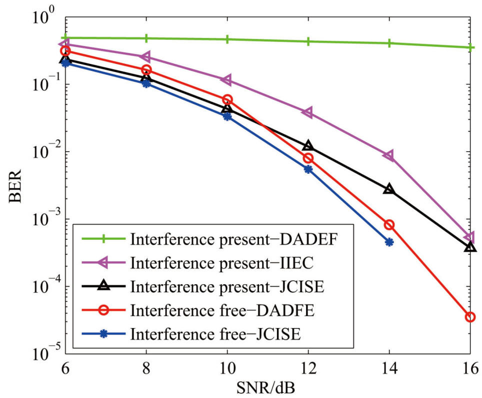

Fig. 4. The BER performance of the several algorithms when SIR = −25 dB in the simulated channels.

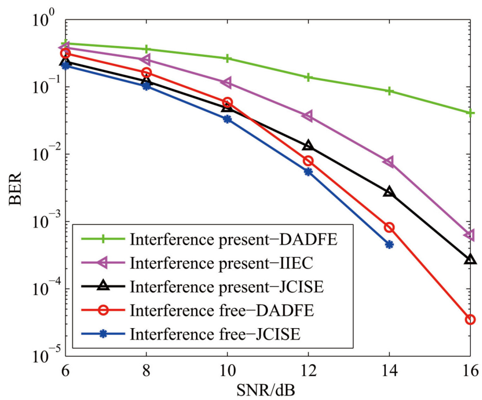

Fig. 5. The BER performance of the several algorithms when SIR = −20 dB in the simulated channels.

and JCISE algorithms are much better than that of conventional DADFE algorithm. The reason is that the ambient noise is assumed Gaussion distribution for DADFE, but this is far from the actual noise distribution in the environment with impulsive interference and abnormal energy fluctuation will cause divergence of adaptive algorithm. Obviously, this situation can be significantly improved with iterative interference estimation and cancellation in IIEC algorithm. And the performance will be further enhanced with joint estimation in JCISE algorithm, of which the curve is more closely follows that of the interference-free scenario. The SIR level has a great impact on the conventional DADFE algorithm. When SIR = −20 dB, the BER curve decreases with increasing SNR. When SIR = −25 dB, the curve is very flat with the changes in the SNR. Fig. 6 shows the BER performance for the three algorithms when SIR = −25 dB in the experimental channel. The result is similar to that in the simulated channels. JCISE algorithm is the most robust, while DADFE algorithm cannot decode effectively due to the interference

VI. Conclusion

This letter develops an two-step equalization algorithm for single-carrier modulated UWA communications. In the

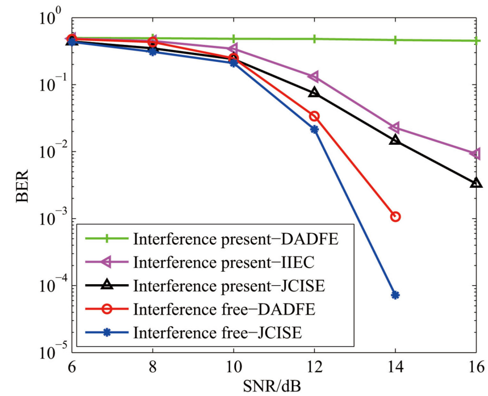

Fig. 6. The BER performance of the several algorithms when SIR = −25 dB in the experimental channel.

first step, the interfering waveform is detected using a Page test detector, and an iterative equalization with interference estimation/cancellation is operated based on the parameterized interference model. In the second step, a joint sparse channel, impulsive interference, and symbols estimation is performed to improve the equalization performance. Simulations results show that the proposed JCISE algorithm can effectively mitigate the partial-band partial-block-duration interference and has significant performance advantages over IIEC and conventional DADFE algorithm in various SIR levels.

References

[1] J. A. Catipovic, "Robust acoustic communication with underwater vehicles in noisy and jammed shallow water environments," in *Proc. Oceans Mastering Oceans Through Technol.*, 1992, pp. 1–9.

[2] Y. Tian, X. Han, J. Yin, Q. Liu, and L. Li, "Group sparse underwater acoustic channel estimation with impulsive noise: Simulation results based on arctic ice cracking noise," *J. Acoust. Soc. Amer.*, vol. 146, no. 4, pp. 2482–2491, 2019.

[3] M. Marey and H. Steendam, "Analysis of the narrowband interference effect on OFDM timing synchronization," *IEEE Trans. Signal Process.*, vol. 55, no. 9, pp. 4558–4566, Sep. 2007.

[4] M. Chitre, S. Kuselan, and V. Pallayil, "Ambient noise imaging in warm shallow waters; robust statistical algorithms and range estimation," *J. Acoust. Soc. Amer.*, vol. 132, no. 2, pp. 838–847, 2012.

[5] M. Chitre, S. Ong, and J. Potter, "Performance of coded OFDM in very shallow water channels and snapping shrimp noise," in *Proc. OCEANS MTS/IEEE*, 2005, pp. 996–1001.

[6] X. Kuai, H. Sun, S. Zhou, and E. Cheng, "Impulsive noise mitigation in underwater acoustic OFDM systems," *IEEE Trans. Veh. Technol.*, vol. 65, no. 10, pp. 8190–8202, Oct. 2016.

[7] P. Chen, Y. Rong, S. Nordholm, Z. He, and A. J. Duncan, "Joint channel estimation and impulsive noise mitigation in underwater acoustic OFDM communication systems," *IEEE Trans. Wireless Commun.*, vol. 16, no. 9, pp. 6165–6178, Sep. 2017.

[8] S. Wang, Z. He, K. Niu, P. Chen, and Y. Rong, "New results on joint channel and impulsive noise estimation and tracking in underwater acoustic OFDM systems," *IEEE Trans. Wireless Commun.*, vol. 19, no. 4, pp. 2601–2612, Apr. 2020.

[9] Z. Wang, S. Zhou, J. Catipovic, and P. Willett, "Parameterized cancellation of partial-band partial-block-duration interference for underwater acoustic OFDM," *IEEE Trans. Signal Process.*, vol. 60, no. 4, pp. 1782–1795, 2011.

[10] W. Ge, Z. Wang, J. Yin, and X. Han, "Parameterized interference cancellation for single-carrier underwater acoustic communications," in *Proc. IEEE Global Oceans U.S. Gulf Coast*, 2020, pp. 1–6.